D.Y. PATIL COLLEGE OF ENGINEERING &TECHNOLOGY,
KASABA BAWADA, KOLHAPUR
DEPARTMENT OF COMPUTER SCIENCE & ENGINEERING
(Academic Year: 2026-2027)



D Y PATIL
COLLEGE of
ENGINEERING & TECHNOLOGY
(AN AUTONOMOUS INSTITUTE)
KASABA BAWADA, KOLHAPUR

APC Report

On

"Job Market Skills Analysis"

Class: TY (CSE) Div.: C

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1. Introduction

In today's digital world, data analysis plays an important role in understanding trends, making decisions, and solving real-world problems. The job market is continuously changing due to technological advancements, digital transformation, and the increasing demand for skilled professionals. Companies require candidates with different technical skills, such as Python, Java, SQL, Machine Learning, Data Analysis, Web Development, Cloud Computing, Cybersecurity, C++, and Power BI. Therefore, understanding the demand for different technical skills and their associated salary trends is important for students, job seekers, and educational institutions.

The Job Market Skills Analysis project is a web-based data analysis application developed using Python and Streamlit. The main purpose of this project is to analyze job-related data and present useful information about technical skills, sample job openings, and estimated average salaries. The project provides a simple and interactive dashboard that allows users to explore the available data and understand the differences in demand for various technical skills.

The system uses a structured dataset containing information about technical skills, job openings, and average salary estimates in lakhs per annum (LPA). Python is used to implement the application logic, while Pandas is used for data manipulation and analysis. NumPy supports numerical operations where required, and Matplotlib is used to create graphical visualizations. Streamlit provides a user-friendly interface through which the analysis results are displayed in the form of tables, charts, statistics, and summaries.

The dashboard includes several important features, such as skill selection filters, a dataset table, a job openings comparison chart, a pie chart showing the distribution of sample job openings, and a chart comparing estimated average salaries. It also displays key statistics, including total sample job openings, average openings per skill, and estimated average salary. These features help users examine the data more easily and compare the information associated with different technical skills.

For example, in the sample dataset used in this project, Python has 850 sample job openings, making it the skill with the highest number of openings among the listed skills. Machine Learning has the highest estimated average salary in the displayed sample, at ₹9.5 LPA. These results demonstrate how data analysis and visualization can be used to identify patterns within a dataset. However, these values are illustrative sample results and should not be considered actual or current job market statistics.

The main objective of this project is to make job-related information easier to understand through graphical representation and interactive data exploration. Instead of manually examining rows of data, users can view charts and summaries to compare skills and salary estimates. The project also provides practical experience in Python programming, data processing, statistical analysis, and dashboard development.

Furthermore, this project demonstrates the practical application of data analysis technologies in a career-related domain. It can help students understand the importance of technical skills and encourage them to explore areas for further learning. The dashboard can also serve as a foundation for more advanced applications involving real-time job postings, location-based analysis, salary comparisons, and machine learning techniques.

In conclusion, the Job Market Skills Analysis project provides a simple and effective way to explore job-related data using an interactive web application. By combining Python, Pandas, Matplotlib, and Streamlit, the system transforms structured data into meaningful visual information. Although the current version uses illustrative sample data, it can be further improved by integrating authentic job market datasets and regularly updated information to provide more realistic insights into industry skill requirements and salary trends.

2. Problem Statement

Students and job seekers often face difficulties identifying which technical skills are in demand and understanding how job opportunities and salaries vary across different skills. Manually examining job-related information can be time-consuming and may make it difficult to compare multiple skills effectively.

Therefore, there is a need for a simple and interactive system that analyzes job market data and presents meaningful information through statistics, tables, and charts.

The proposed Job Market Skills Analysis system addresses this problem by processing a structured dataset, calculating job-opening and salary statistics, and displaying the results through an interactive dashboard. It helps users explore the available data and understand the differences in skill demand and estimated salaries.

3. Objectives

The main objectives of the Job Market Skills Analysis project are:

1. To analyze job market data and identify the demand for different technical skills.
2. To calculate and compare the number of job openings associated with various skills.
3. To analyze and compare estimated average salaries for different technical skills.
4. To present job market information using interactive tables, charts, and graphs.
5. To provide a user-friendly dashboard for exploring job-related information.
6. To help students and job seekers understand skill requirements and plan their learning accordingly.

4. Technologies Used

The project uses the following technologies and tools:

Technology	**Purpose**
Python	Main programming language used for data processing and application logic.
Streamlit	Used to develop the interactive web-based dashboard.
Pandas	Used to load, organize, filter, and analyze tabular data.
NumPy	Used for numerical calculations and data operations, if required by the implementation.

Matplotlib	Used to create graphical visualizations, including bar charts and pie charts.
CSV Dataset	Used to store the sample job market data, if the project loads data from a CSV file.
Web Browser	Used to access and interact with the dashboard.

5. System Architecture

The system follows a simple data analysis architecture. The dataset is provided to the Python application, where the data is processed and analyzed. Streamlit presents the resulting statistics and visualizations through a web-based dashboard.

System Architecture Diagram:

Dataset → Data Loading → Data Processing and Analysis → Statistical Calculations → Data Visualization → Streamlit Dashboard → User

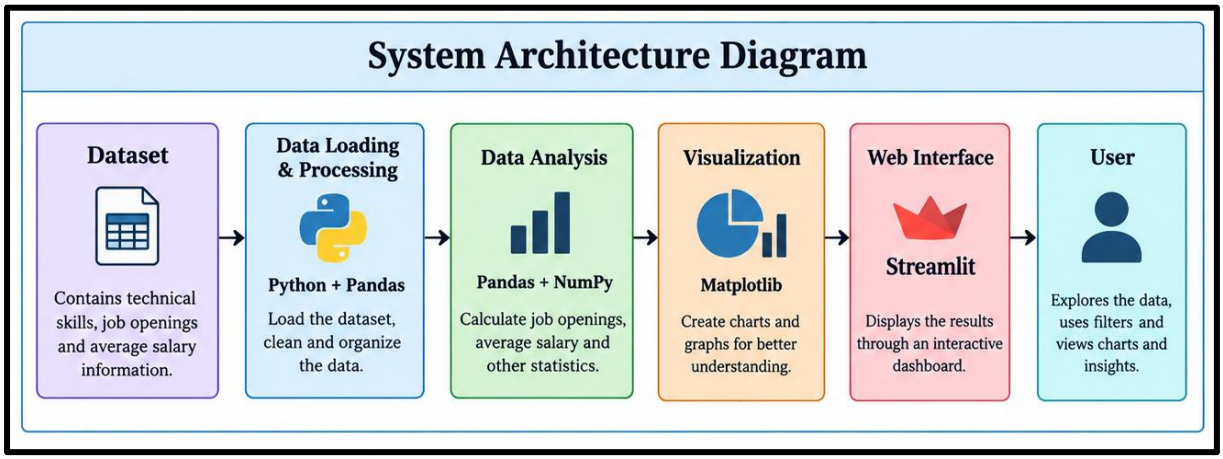


Fig 1: System Architecture

Components of the Architecture:

1. Data Source: Contains technical skills, job openings, and estimated average salary information.
2. Data Processing: Python and Pandas are used to load and organize the dataset.
3. Analysis Module: Calculates job-opening statistics and compares estimated salaries.
4. Visualization Module: Matplotlib generates graphs and charts to represent the analysis.
5. User Interface: Streamlit displays statistics, tables, filters, charts, and summaries.
6. User Interaction: Users select skills and explore the available results through the dashboard.

6. Working of the System

The working of the Job Market Skills Analysis system is explained in the following steps:

Step 1: Dataset Loading

The system loads a structured dataset containing information about technical skills, job openings, and estimated average salaries.

Step 2: Data Processing

Python processes the dataset, and Pandas is used to organize and manipulate the data for analysis.

Step 3: Skill Selection

Users can select technical skills such as Python, Java, SQL, Machine Learning, Data Analysis, and Web Development using the interactive filter.

Step 4: Statistical Analysis

The system calculates statistics such as total sample job openings, average openings per skill, and estimated average salary.

Step 5: Data Visualization

The analyzed information is displayed through tables, bar charts, and a pie chart. These visualizations help users compare skill demand and salary estimates.

Step 6: Summary Generation

The dashboard highlights key findings, such as the skill with the highest number of sample openings and the skill with the highest estimated average salary.

Step 7: Result Display

The final results are displayed on the Streamlit dashboard, allowing users to explore and interpret the data easily.

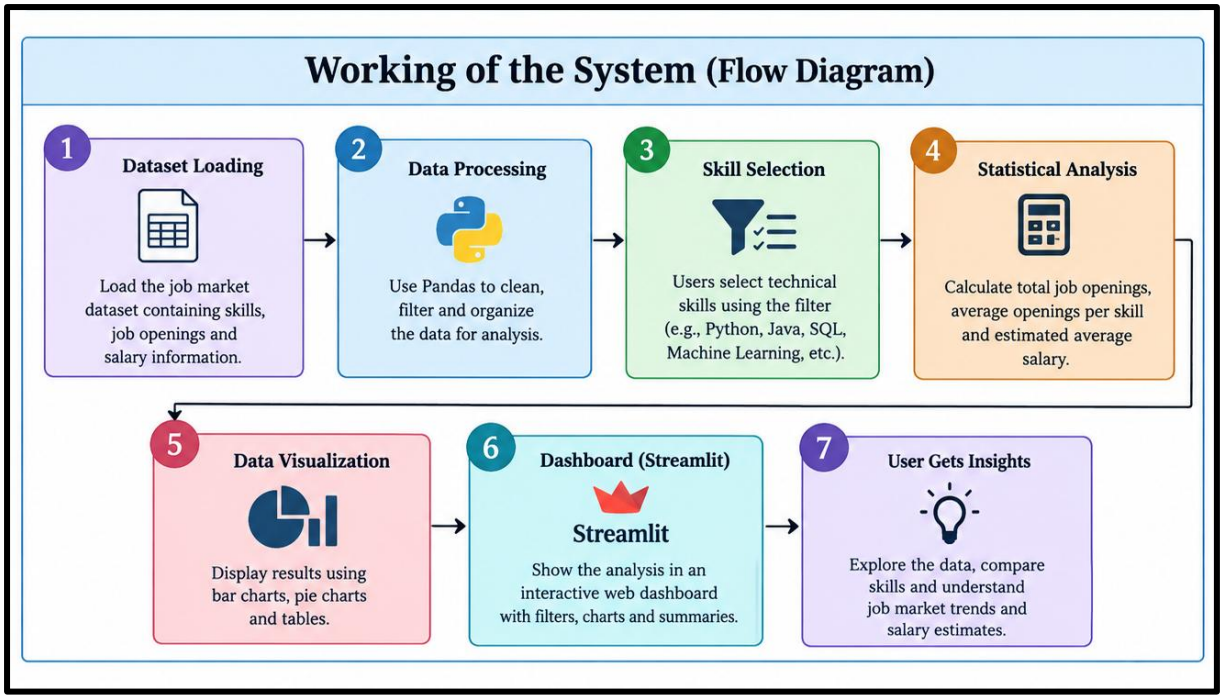


Fig 2: Working Flow of the System

7. Main Features

The main features of the project are:

1. Interactive Dashboard: Displays job market analysis in a clear and organized format.
2. Skill Filters: Allows users to select and explore specific technical skills.
3. Statistical Overview: Displays total sample job openings, average openings per skill, and estimated average salary.
4. Dataset Display: Presents the job market data in a tabular format.
5. Job Openings Chart: Compares the number of sample job openings across different skills.
6. Opening Distribution Chart: Displays the distribution of sample job openings using a pie chart.
7. Salary Comparison Chart: Compares the estimated average salaries associated with different skills.
8. Summary Section: Highlights important findings from the dataset.
9. Web-Based Access: Allows users to access the application through a compatible web browser.

8. Advantages and Limitations

8.1 Advantages

1. Provides an easy-to-use interface for analyzing job market data.
2. Helps compare the demand for different technical skills.
3. Presents information through graphical visualizations.
4. Makes salary comparisons easier to understand.
5. Helps students identify skills they may wish to learn.
6. Reduces the effort required to examine tabular data manually.
7. Can be extended with larger datasets and additional analysis features.

8.2 Limitations

1. The current dashboard uses illustrative sample data rather than live job market statistics.
2. The accuracy of the analysis depends on the quality of the dataset.
3. The results may not represent the entire job market.
4. Salary estimates may vary according to experience, location, company, and job role.
5. The current system does not guarantee employment or predict individual career success.
6. Real-time job listings and automatic data collection are not included unless implemented separately.

9. Applications

The Job Market Skills Analysis project can be useful in the following areas:

1. Student Career Planning: Helps students explore the demand for technical skills.
2. Job Seekers: Helps users compare skill demand and estimated salary information.
3. Educational Institutions: Can support discussions about industry-relevant technical skills.
4. Skill Development: Helps learners identify areas for further study.
5. Career Guidance: Can provide supporting information for career counselling.
6. Data Analysis Education: Demonstrates the practical use of Python, Pandas, and data visualization.
7. Recruitment Research: Can support preliminary analysis of job-related datasets.

10. Conclusion

The Job Market Skills Analysis project demonstrates how Python and Streamlit can be used to develop an interactive data analysis application. The system processes job market data and presents information about technical skills, sample job openings, and estimated average salaries through statistics, tables, and charts.

The dashboard makes it easier to compare different skills and understand patterns in the available dataset. It also demonstrates the practical application of data processing and visualization techniques.

Although the current implementation uses illustrative sample data, the system can be extended with authentic job postings and regularly updated datasets to provide more useful insights into changing job market requirements.

11. Future Scope

The project can be improved in the following ways:

1. Real-Time Job Data: Integrate authorized APIs or other permitted data sources to obtain current job postings.

2. Location-Based Analysis: Compare job openings across different cities, states, and countries.

3. Advanced Salary Analysis: Compare salaries based on experience, job role, company, and location.

4. Machine Learning Integration: Develop models to investigate future skill-demand trends when suitable historical data is available.

5. Personalized Skill Recommendations: Suggest relevant skills based on a user's interests and career goals.

6. Additional Filters: Add filters for job roles, experience levels, locations, and salary ranges.

7. Database Integration: Store and manage larger datasets using a suitable database if required.

8. Improved Dashboard: Add more visualizations, downloadable reports, and additional interactive features.

12. References

1. Python Official Documentation — https://docs.python.org/3/

2. Streamlit Documentation — https://docs.streamlit.io/

3. Pandas Documentation — https://pandas.pydata.org/docs/

4. NumPy Documentation — https://numpy.org/doc/

5. Matplotlib Documentation — https://matplotlib.org/stable/

13. Implementation Screenshots

The following screenshots demonstrate the implementation and output of the Job Market Skills Analysis project.

13.1 Login Page



Fig. 1: Login Page

13.2 Job Market Skills Analysis Dashboard

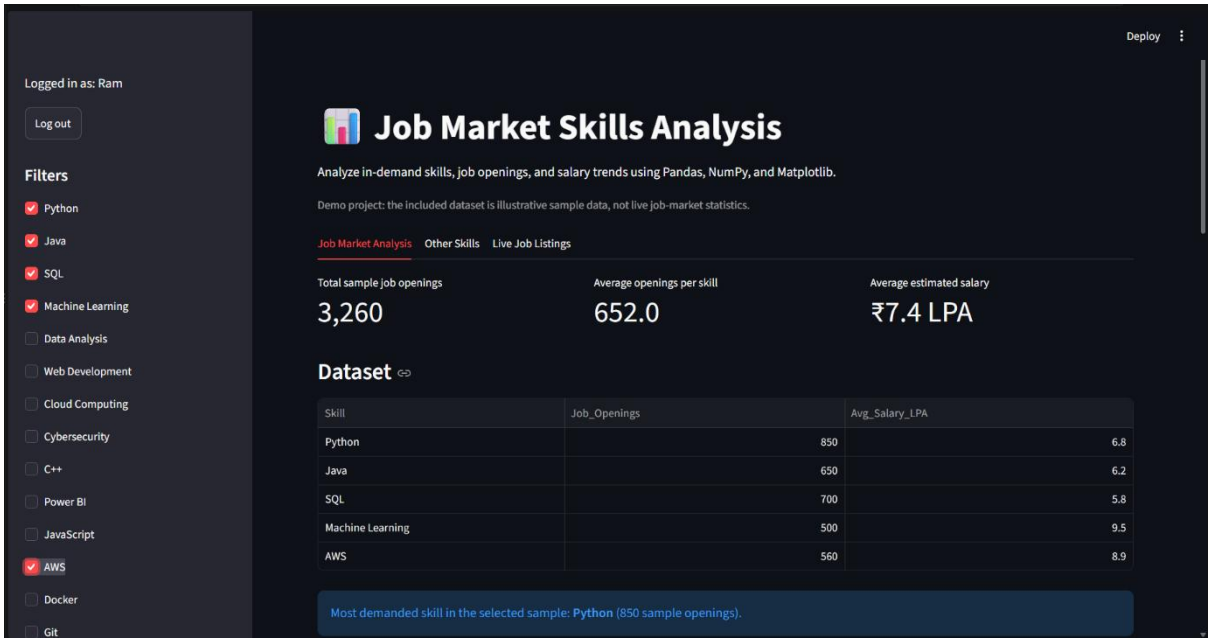


Fig. 2: Job Market Skills Analysis Dashboar

13.3 Job Openings by Skill

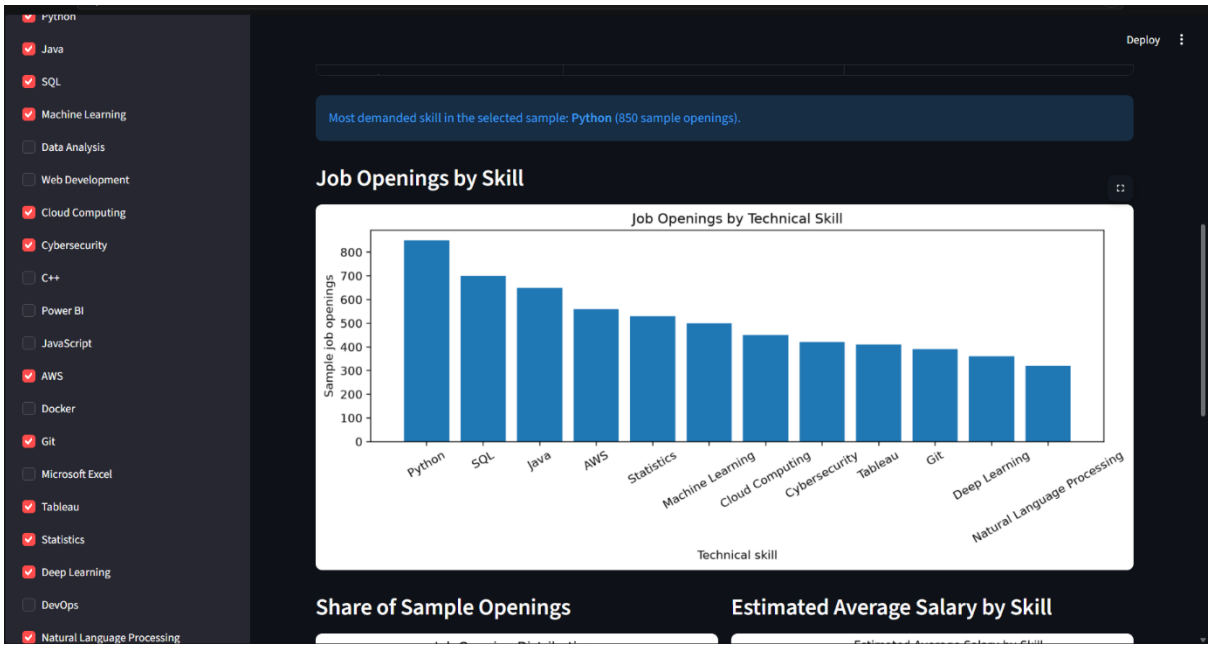


Fig. 3: Job Openings by Skill

13.4 Estimated Average Salary by Skill

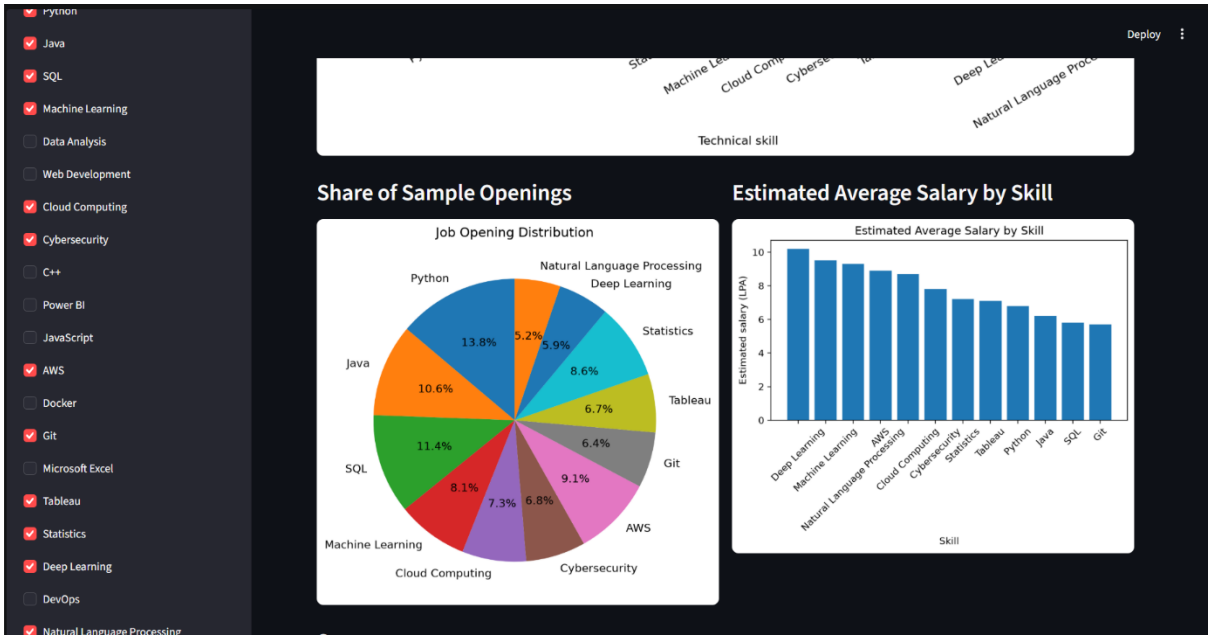


Fig. 4: Estimated Average Salary by Skill

13.5 Live Job Listings

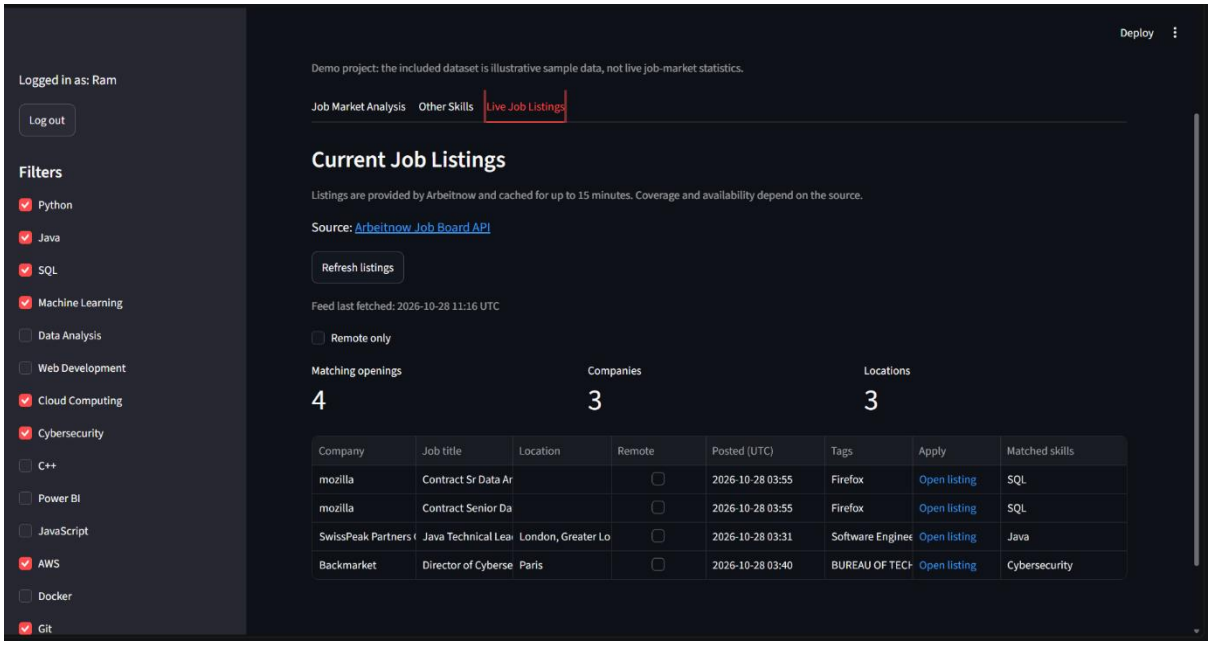


Fig 5: Live Job Listings